Buck converter

IC: Diodes inc. AP62300T

$I_{max} = 3A$, $f_{sw} = 750KHz$

V_{in}: 4V → 18V (20V max)

2 internal mosfets

75mΩ high, 45 mΩ low

ref $V_{out} = 0.763V$,

V_{out}: 0.8V → 7V

Inductor:

$$D = \frac{V_{out}}{V_{in\,max}} = \frac{5V}{13.5V} = 0.37$$

$V_{in} = 13.5V$ $V_{out} = 5V$ $I = 2A$ (or 3A?)

$$L = \frac{V_{out}(V_{in} - V_{out})}{V_{in} \cdot \Delta I_L \cdot f_{sw}}$$

ΔI_L = inductor current ripple

= 0.2 - 0.3 of max I @ 3A.

= 2A · 0.3 = 0.9A

$$= \frac{5(13.5-5)}{13.5 \cdot 0.9 \cdot 750K} = \frac{42.5}{911250} \approx 4.66\mu H$$

use 4.7 µH, closest standard value

$$I_{peak} = I + \frac{\Delta I_L}{2} = 2 + \frac{0.9}{2} = 2.45A$$

$I_{sat} > 1.35 \times I_{max} = 1.35 \cdot 3$

$I_{sat} > 4.05$

$R_I < 30m\Omega$

Buck IC data sheet

$I_{sat} > 4.05 + \Delta I_L$

$I_{sat} > 4.05 + 0.9A$

$I_{sat} > 5A$

Table 1. Recommended Component Selections

AP62300/AP62301/AP62300T								
Output Voltage (V)	AP62300/AP62301 R1 (kΩ)	AP62300T R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C5 (pF) (Optional)	C6 (nF)
1.2	4.99	5.76	10	1.5	10	2 x 22	Open	100
1.5	8.66	9.76	10	1.5	10	2 x 22	Open	100
1.8	12.4	13.7	10	2.2	10	2 x 22	10 - 100	100
2.5	21.5	22.6	10	2.2	10	2 x 22	10 - 100	100 - 220
3.3	31.6	33.2	10	3.3	10	2 x 22	10 - 100	100 - 330
5.0	52.3	56.2	10	3.3	10	2 x 22	10 - 100	100 - 330

Input caps:

- $I_{in} = I_{out} \cdot \sqrt{D \cdot (1-D)} = 2A \cdot \sqrt{0.37 \cdot 0.63} = 0.97A$ (worst case 1A @ D=0.5)
- From IC datasheet, RMS > $\frac{1}{2} I_{max}$, 2 capacitors give 0.48A each
- Input ripple $= I_{out} \cdot D \cdot \frac{(1-D)}{(f_{sw} \cdot C_{in})} = \frac{0.466}{750K C_{in}}$, with ~ 0.5-0.65x capacitance @ 13.5V

  $C_{in} \approx 10-13\mu F$, ripple $\approx 48-62mV = 0.4\% \cdot 13.5V$
- 25V rating on 15V bus is 1.67x, > 50% margain
- using 1206 to keep more capacitance under DC bias than 0603

Bootstrap cap: datasheet uses 100nF, allows for 100-330nF

Output Caps

$$\Delta I_L \cdot \left(ESR + \frac{1}{8 \cdot f_{sw} \cdot C_{out}}\right) = \text{output ripple}$$

$C_{out} \approx 17.6 - 26.4\mu F$, $ESR = 5m\Omega$

ripple $= 0.95(.005 + .0095) = 14mV \rightarrow 0.95(.005 + .0063) = 11mV$

2a w/ 0.25V, overshoot $C_{out} > L \cdot \frac{I^2}{(\Delta V \cdot V_{out})} = 4.7\mu \cdot \frac{4}{(0.25 \cdot 5)} = 15\mu F$

undershoot $C_{out} > L \cdot \frac{I^2}{\Delta V \cdot (V_{in} - V_{out})} = 4.7\mu \cdot \frac{4}{0.25 \cdot 8.5} = 8.9\mu F$

C_2 from data sheet is also 2 x 22µF caps

Resistors

$$R_1 = R_2 \left(\frac{V_{out}}{V_{fb}} - 1\right) = 10K \left(\frac{5}{0.763} - 1\right) = 55.5K$$

not standard, $R_1 = 56.2K$

$V_{out} = .763\left(1 + \frac{56.2}{10}\right) = 5.05V$, $I = \frac{5.05}{66.2K} = 76\mu A$

worst case V_{out} is 4.86 → 5.24V (0.747 - 0.778 over temp + 1% resistors)

ie, 5V ± 3.37%

R_3 / R_4 divider

$V_{on} = 9V$, $V_{off} = 7V$ EN pin $I_{off} = 1.5\mu A$, $I_{on} = 7\mu A$

$R_3 = (0.917 \cdot V_{on} - V_{off}) / 5.625\,\mu A = 223K$ → using standard val of 221kΩ

$R_4 = 1.1 \frac{R3}{V_{off} - 1.1 + 7\mu A R_3} = 32.9K$ → using standard val of 32.4kΩ

$k = 1 + 221/32.4 = 7.82$

$V_{on} = 1.2 \cdot 7.82 - 1.5\mu A \cdot 221k\Omega = 9.05V$

$V_{off} = 1.1 \cdot 7.82 - 7\mu A \cdot 221K = 7.06V$

at worst case 1.25V, $V_{on} = 9.44V$, < 10V buck min

13.5V = 53mA, V_{EN} @ 15V = 2V, 2V < 8V abs. max